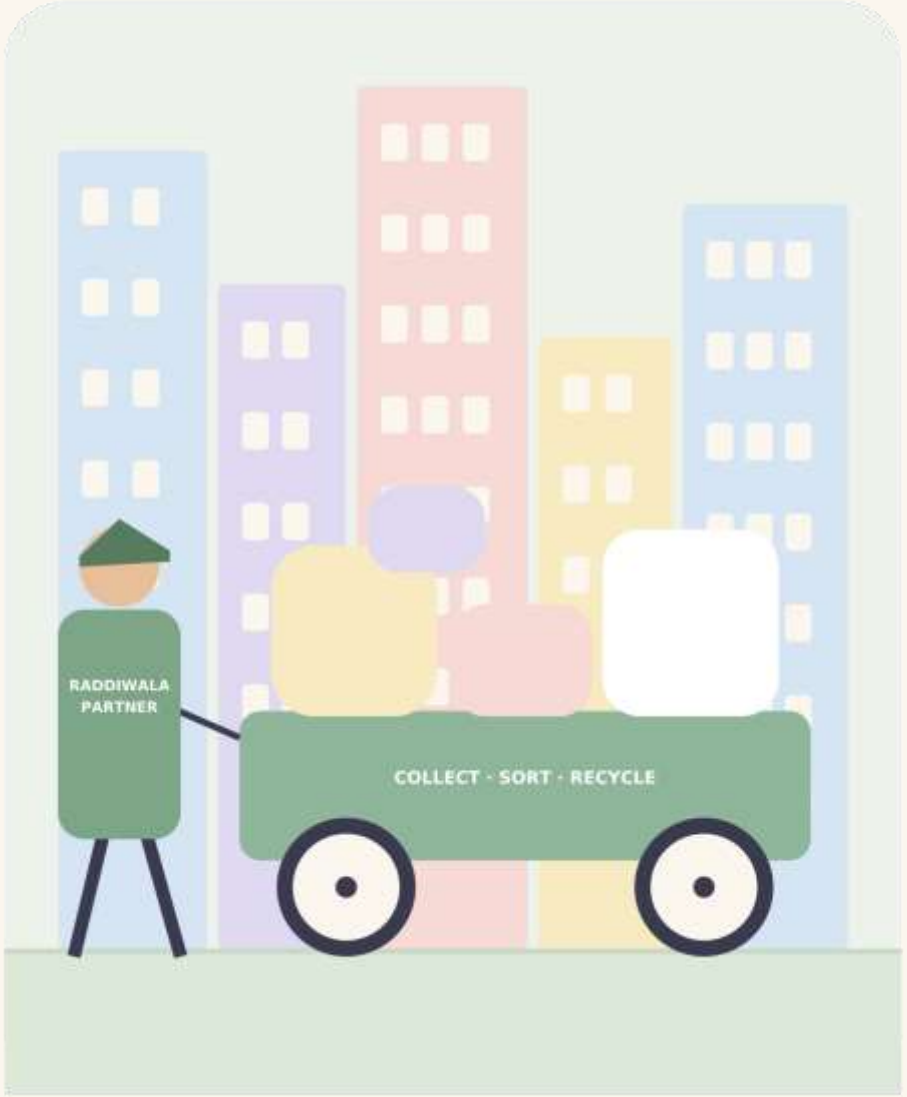
MITSUI & CO. HACKATHON · MITSUI-PS-01

Building Collaborative Circular Waste Management Systems in Indian Cities

circular. by Terra — Hybrid app + WhatsApp coordination connecting households, societies, shops, volunteer farmers, informal collectors and recycling hubs. Traceable streams, powered by AI segregation & verified movement.

Source segregation → Informal collection & fair pay → Material traceability → Circular recycling

WET DRY RESIDUAL



Segregation incentives

Verified material movement

Dignified collector livelihoods

Source segregation · Collection & fair pay · Traceability · Circular recycling

Problem Understanding & Local Relevance

WHAT GOES WRONG ON THE GROUND

1 Mixed waste at the source

Wet, dry and sanitary waste leave the home together, so recyclable material gets soiled and ends up in landfill.

CPCB 2020–21: over 1.6 lakh tonnes of municipal waste a day, with over a quarter unaccounted for.

2 Informal collectors are invisible

Kabadiwalas and raddiwalas have no records, no price visibility and no digital income trail.

ILO / Chintan: 1.5–4 million informal pickers handle about 60–70% of urban recyclables.

3 No trust in the chain

Shops and households cannot see where material goes, or whether the weight they handed over is the weight paid for.

India's EPR rules largely overlook informal pickers (Down To Earth).

WHY THIS FITS INDIAN CITIES

Everyone already uses WhatsApp

Collectors need no new app, just the phone they already carry.

Raddiwalas become partners

The existing informal network is strengthened, not replaced.

Two housing realities

Independent houses and gated societies each get a pickup model that suits them.

Pioneer ward

Start with a few societies, one market and nearby collectors.

WHO DOES WHAT

Households & societies

Segregate at source and earn rewards.

Informal collectors

Collect, weigh and get paid on verified weight.

Municipality

Gets ward-level data; handles residual and sanitary waste.

Shops & businesses

Segregate, sell scrap and get paid.

Corporations & industries

Buy verified material and EPR credits.

Sources: CPCB Annual Report 2020–21 (via Civil Services Chronicle); ILO and Chintan estimates and EPR commentary (via Down To Earth).

Homes, Societies, Shops & Farms

Individual houses

Direct doorstep pickup with regular collection enrollment. Segregated waste is collected free.



Large societies

If the society collaborates, maids / househelps drop segregated waste at a collection unit inside the premises during a fixed time window.



Shops & local markets

Tailors give cloth scraps; vegetable and juice vendors get free wet-organic pickup; businesses are paid for metal and textile scrap.



FREE **Segregated waste**
Free regular pickup

FEE **Non-segregated waste**
Fee applies; sorted at secondary hub

01



Biodegradable

Community volunteer drop-off stations & local farm hosts feeding Hub Silo #8 for municipal composting.

02



Non-biodegradable

Segregation units, recycling plants or second-hand markets, incl. torn clothing and old bags by condition

03



Paper / cardboard

Raddiwala and paper mills

VOLUNTEER HOSTS

Farmers & volunteer community stations

Sign up in the app as an organic drop-off point. Take wet waste from neighbors & shops; compost locally or route to BioCompost Silo #8. Relieves municipal burden.

CLOSED LOOPS · WHERE EACH STREAM ENDS UP

- **Organics** → biogas & compost
- **Plastics** → authorized recyclers
- **Paper & cardboard** → paper mills
- **Metal** → scrap dealers & foundries
- **E-waste** → authorized processors
- **Textiles** → reuse or second-hand

App Features & QR Traceability

1 Dispatch

Collector assigned by 6-digit postal PIN; receives WhatsApp alert with pickup list.

2 QR link

Collector scans doorstep QR or enters 4-digit PIN to instantly verify household handover.

3 Doorstep AI check & weigh-in

Terra AI camera checks contamination; collector logs initial Weight 1 on digital scale.

4 Dual-scale hub audit

Hub secondary scale logs Weight 2; >10% variance triggers automatic audit alert.

Households, societies & shops

Easy to use, with habit-building nudges

Booking & shop QR

Doorstep or on-demand pickup, with every handover recorded.

"How to Segregate" guide

Interactive and step by step for each waste stream.

Points & leaderboard

Points for verified segregation, plus a society and ward leaderboard.

Terra AI scanner & dual home

Collectors, on WhatsApp

No new app to install

WhatsApp bot & voice notes

Verified collector profile

Earnings history and a visible rate card give formal recognition.

Instant digital payout

Paid on verified weight, with a clear earnings view.

Stream-wise routing

Keep wet, dry and paper separate in the cart.

Hubs, hosts, plants & industries

Control, visibility and markets

Dashboard & EPR reports

Collections by material, collector and ward, for hubs and the municipality.

Volunteer station & Silo #8

Recycling marketplace

Aggregators list verified lots; industries buy at transparent prices.

Society collection slots

Set the drop-off unit and time window for househelps.

Hybrid by design: an app for households, shops, hosts and industries; WhatsApp for collectors; a web dashboard for hubs, plants and the municipality.

Technical Architecture & Prototype

CHANNELS

- Household, shop & host app
- Collector on WhatsApp
- Hub, plant & industry dashboard

↓

Direct WhatsApp WebSockets (Baileys) · Groq Whisper LPU · Event Stream

↓

BACKEND SERVICES

- Dispatch & routing
- QR & bag registry
- Weight reconciliation
- Payout engine
- Recycling marketplace
- Notifications

↓

DATA

- Immutable event ledger
- Operational database
- Analytics & EPR reports

TECHNICAL DEPTH

Data model
Shop, bag, pickup, weigh-in, lot and payout records, linked by IDs.

Pricing
Recycling units publish a rate card per material; payout = verified weight × rate.

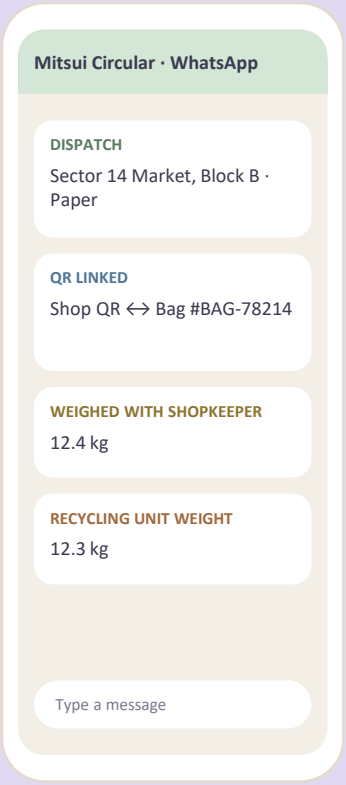
Micro-depot routing
Pickups clustered by ward and stream, sent to the nearest depot or host.

Dual-scale audit protection
Two independent weighings (Doorstep W1 vs Hub Scale W2); >10% variance triggers fraud flag.

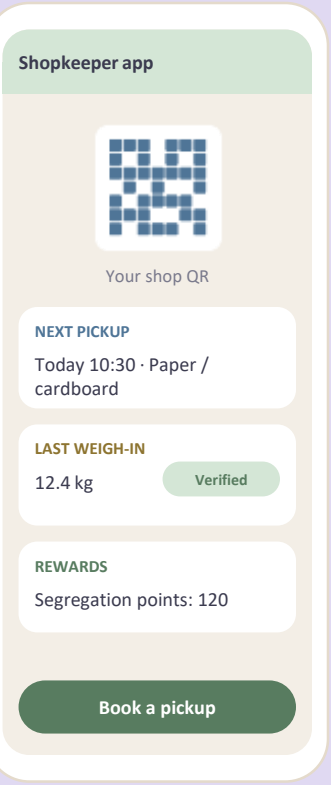
Voice accessibility
Groq Whisper sub-300ms Hindi/Hinglish voice notes for field collectors on WhatsApp.

Privacy & consent
Personal data kept with user consent; buyers see only shop IDs.

PROTOTYPE SCREENS · ILLUSTRATIVE



Collector view



Shopkeeper view

Source segregation · Collection & fair pay · Traceability · Circular recycling

Feasibility, Impact & Scalability

UNIT ECONOMICS · PER ₹1,000 OF VERIFIED MATERIAL

40% — **Waste-generating business**
₹400 · rewards source segregation

40% — **Collector / kabadiwala**
₹400 · digital payout on verified weight

20% — **Our platform**
₹200 · logistics, QR traceability, free household pickup, EPR credit sales via eligible partners

Illustrative: a collector netting ₹300 on ₹1,000 of material today would receive ₹400 here (about +33%), from better prices, cleaner loads and fewer wasted trips.

ROLLOUT

Phase 1 · Pilot
One ward: a few societies, one market, existing raddiwalas and a couple of farm hosts.

→

Phase 2 · Expand
More societies and shops, hub and municipal partnerships, recycling-unit weighing at scale.

→

Phase 3 · City-wide
Full coverage with EPR-ready reporting and credit workflows.

Scales ward by ward, city by city and stream by stream on one WhatsApp + QR stack, wherever informal collectors already operate.